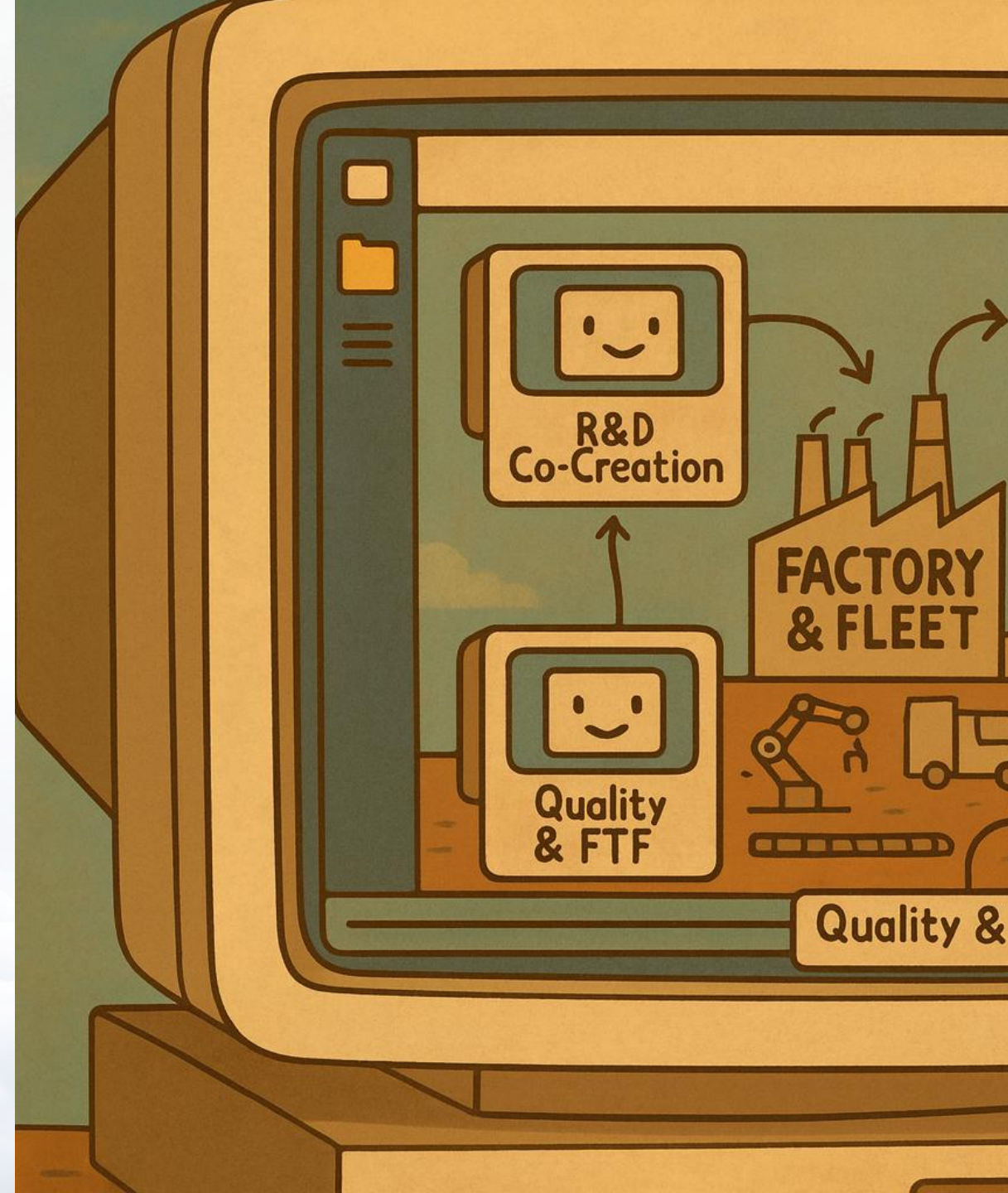




Manufacturing: Apps that matter

Disclaimer: This content is for Microsoft partner guidance only and not to be shared directly with customers.



What is an AI frontier firm

→ The Frontier Firm Defined

A next-generation company that fuses human judgment with AI agents to scale faster, work smarter, and unlock new value - powered by on-demand intelligence and human + agent hybrid teams.

💡 Core Characteristics

- **AI-operated, human-led:** Agents execute tasks; humans provide oversight.
- **Work Chart vs Org Chart:** Teams form around outcomes, not departments.
- **Agent boss mindset:** Every employee manages and collaborates with AI agents.

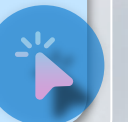
L Why This Matters Now

- **82%** of leaders say 2025 is a pivotal year to rethink strategy and operations.*
- **80%** expect agents to be integrated into their AI strategy within 12–18 months.*
- Frontier Firms outperform peers in productivity, innovation, and speed to value

"As AI democratizes access to expertise and intelligence, we'll see the rise of Intelligence Resources departments -much like how HR and IT evolved into core functions. These new divisions will be essential for managing the interplay between humans and AI agents, emerging as a critical source of competitive advantage in the AI-enabled enterprise."

-Karim R. Lakhani, Chair of the Digital, Data, and Design Institute at Harvard,

[Microsoft Work Trends Index Report](#)



AI Apps that matter:

Build AI Apps that drive Frontier Transformation

Business Impact

Delivers clear, measurable value by solving a real problem aligned with user needs and organizational goals: high-value use cases

Innovative

Leverages cutting-edge agentic technologies and real time data to solve complex problems in ways that are innovative, adaptive, and often predictive or autonomous

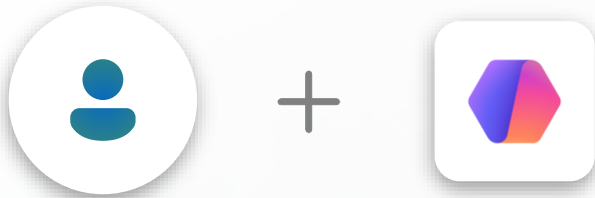
Enterprise-grade

Production-grade systems that are governed, secure, trusted, operationally stable, business-aligned, continuously optimized

Journey to the Frontier Firm

Phase 1

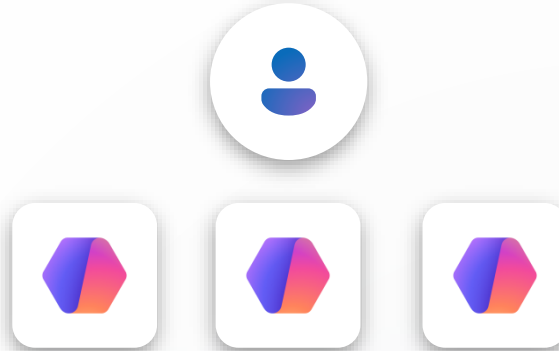
Human with assistant



Every employee has an AI assistant that helps them work better and faster

Phase 2

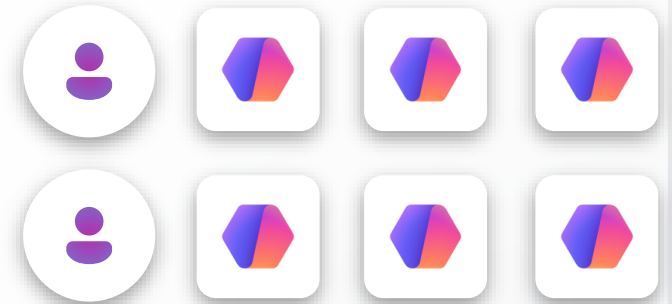
Human-led agents



Agents join teams as "digital colleagues," taking on specific tasks at human direction

Phase 3

Human-led, agent-operated



Humans set direction and agents run entire business processes and workflows, checking in as needed

Becoming Frontier

Success framework

Enrich employee experiences

Reinvent customer engagement

Reshape business processes

Bend the curve on innovation



Approach



Adopt Microsoft copilot



Build in Copilot Studio



Build on Microsoft Azure



Focus of this deck

Guardrails on how to work with the suggested processes

See the whole process



Map the end-to-end flow, actors, policies, and desired outcomes

Call out where agents could take actions

Decompose into "agent-able" jobs



Prioritize by value $\times$ feasibility $\times$ data readiness

Define success metrics and guardrails; timebox the first increment

Select the starting "slice"



Break into sub-processes with clear inputs, decisions, actions, outputs

Create a data map: sources, owners, contracts, latency, quality

Build an enterprise-ready agent



Design for extensibility: APIs/events, modular tools, config over code

Safety & compliance: identity/permissions, policy/RAG, tracing

Scale along the chain



Add/compose agents, automate handoffs, integrate with systems of record

Track ROI; harden to repeatable IP and reuse components

Ground rules: Agentic processes are a combination of multiple software solutions

Manufacturing & Mobility frontier solutions

Overall industry specifics:

- Asset- & safety-critical ops with heterogeneous OT/IT and data silos; strict traceability & security
- High variant complexity; products increasingly software-defined (OTA, cyber)
- Margins in aftermarket/dealer & fleet ecosystems; heavy regulation & sustainability pressure amid skilled-labor gaps

1	Agentic co-innovation	Orchestrate trend mining, patent scans, simulations, and supplier feedback with multi-agents that draft concept briefs, reuse validated components, and tighten stage-gates
2	Pred. maintenance & asset orchestration	Let autonomous agents read IoT streams, predict failures, schedule windows, and trigger parts procurement across ERP/MES/suppliers - cutting downtime and maintenance while keeping fleets and lines available
3	Knowledge mgmt. and preservation	Unify scattered docs, code, CAD/PLM, and service manuals into a governed, searchable hub; use chat over your corpus (incl. legacy systems) to preserve institutional know-how (beyond retiring experts)
4	Factory ops with real-time data	Agents coordinate MES/APS/WMS to keep takt and flow—balancing lines, planning changeovers, kitting materials, and pacing energy use—reducing buffers and idle time while lifting OEE and throughput
5	Engineering-to-order / requirements eng.	Agents mine customer, regulatory, and field signals to draft and trace requirements, generate specs/tests, run DFX checks, and flag change impacts—shortening cycles and reducing rework

Key metrics relevant to the industry

- **Operations & quality:**
Throughput, unplanned downtime, first-pass yield & scrap/rework
- **Supply chain & logistics:**
OTIF/OTD & lead time, forecast accuracy, inventory turns & logistics cost/unit
- **Mobility & customer:**
Fleet availability & cost per mile, warranty/recall & OTA success, retention/NPS & service revenue/CLV



R&D and agentic co-innovation

1. Agentic co-innovation

This case streamlines one core research and development process: turning a market need into a vetted concept brief ready for the first stage-gate. Multi-agents mine trends and patents, reuse validated components, run quick simulations, and collect supplier feedback to produce an evidence-backed brief with risks, costs, and next steps. Teams compress time-to-decision, avoid late rework, and raise the hit rate of new product introductions.

What - Process overview



Current challenges

- Research is scattered across reports, patent sites, and internal wikis; weak signals are missed and duplicates slip through
- Requirements are vague; component teams start computer-aided design work before customer, regulatory, and manufacturing constraints are aligned.
- Knowledge reuse is poor; engineers recreate variants instead of using proven parts and reference designs
- Supplier input and cost signals arrive late, forcing redesign when materials, tooling, or lead times do not fit
- Early simulations are ad-hoc; results are hard to compare and often lack traceability for the stage-gate review



Frontier solution

Process description

- Capture the need (customer problem, target segment, value proposition) and auto-expand with market trends, and competitor benchmarks
- Scan patents and standards with grounded citations; extract claims, freedom-to-operate risks, and design implications
- Propose concept options that maximize reuse of validated parts; auto-generate a concept brief with requirements, targets, and risks
- Trigger lightweight simulations (for example, structural strength, thermal behavior, manufacturability) and cost/lead-time estimates
- Solicit supplier feedback on materials, tooling, and feasibility; incorporate alternatives and trade-offs
- Compile a traceable stage-gate package with evidence, open questions, and a recommended decision (go, refine, or stop)

Solution description

Agents collaborate via events, share context, and hand off to humans for final sign-off:

- **Trend & Patent Scout Agent** mines market sources, patents, and standards; highlights opportunities, risks, and white spaces
- **Concept Drafting Agent** assembles requirements, proposes options that reuse validated components, and generates the concept brief and visuals
- **Simulation & Cost Agent** runs tool-based checks (finite element analysis, thermal analysis, manufacturability, and cost/lead-time estimation) and summarizes results
- **Supplier Co-design Agent** gathers feasibility input, alternates materials or processes, and records commercial constraints
- **Stage-gate Orchestrator** ensures evidence completeness, applies policy and safety

Why – Impact¹



Business Outcomes²

Cost savings:

- 25–40% faster time from idea to approved concept brief through automated research and drafting
- 15–30% fewer late redesigns by integrating supplier, regulatory, and manufacturability checks up front

Growth

- +5–10 percentage-points higher launch success from evidence-based gating and early risk removal.
- Earlier market entry (typically 4–8 weeks sooner for targeted components), enabling incremental revenue in the current model year



Roles addressed

- Chief engineer or head of research and development.
- Program manager or component design lead.

How – Underlying foundation¹

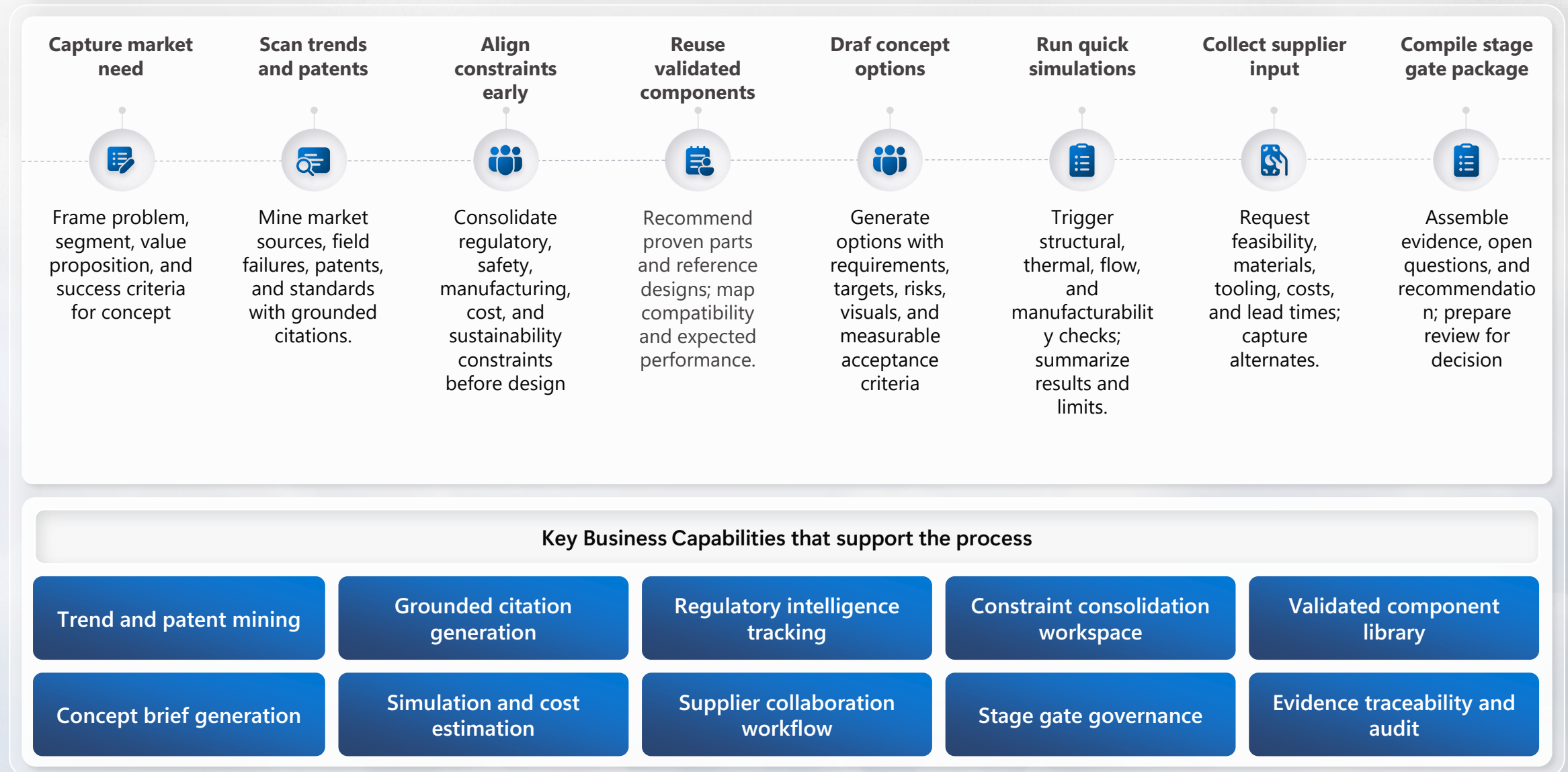
Data to integrate

- Historical designs, validated parts libraries, test reports, and warranty
- Patent and standards databases, competitor specifications, and market trend sources
- Supplier capability, material properties, tooling options, and lead times

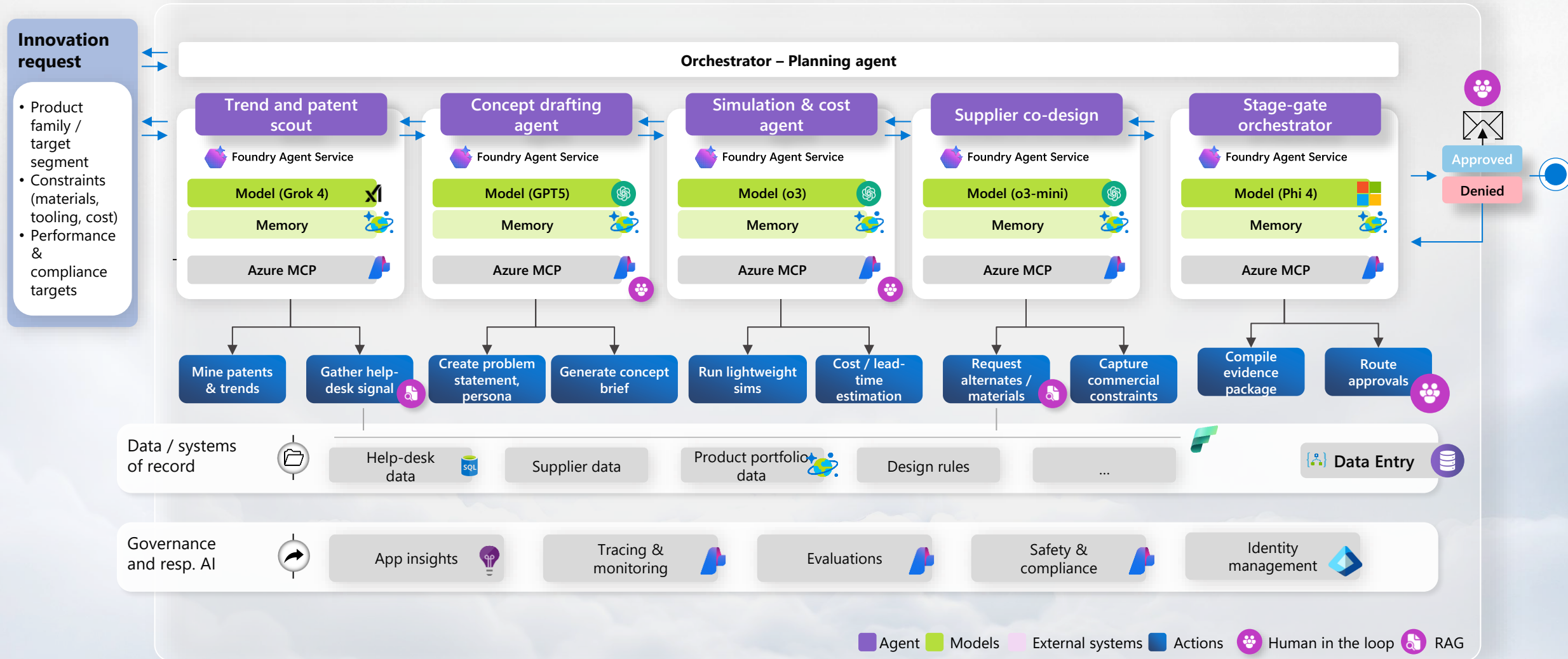
Technical patterns

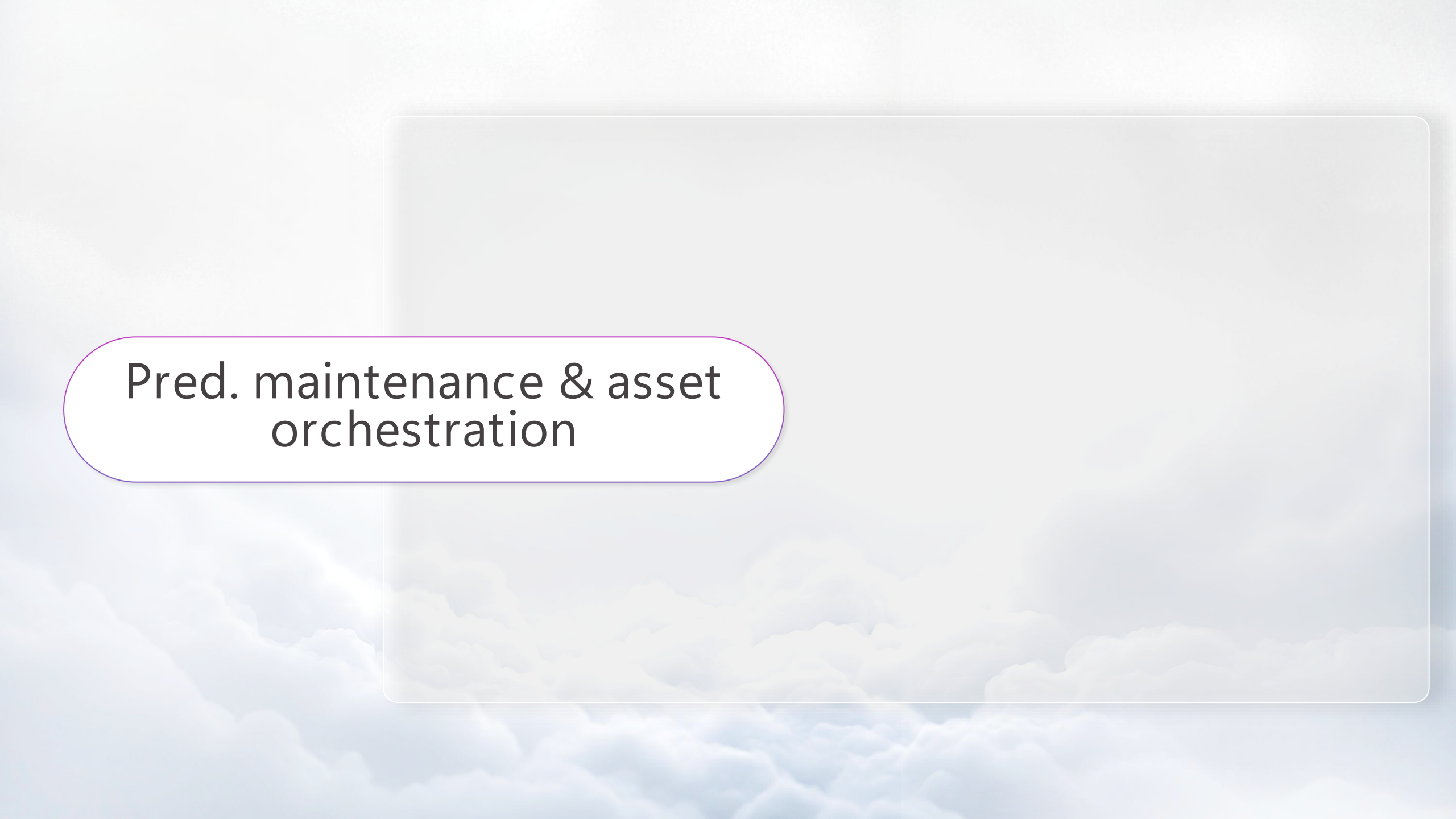
- Multimodal retrieval and synthesis with grounded citations over designs, patents, standards, and field data to drive requirement and concept generation
- Tool-using agent orchestration for simulations and cost estimation with event-driven workflows and human-in-the-loop stage-gate governance

1. What does the innovation and R&D processes look like?



1. Example agentic product innovation





Pred. maintenance & asset
orchestration

2. Predictive maintenance & asset orchestration

Predictive maintenance and asset orchestration let autonomous agents read live Internet of Things signals, predict failures, and coordinate the work to keep equipment, lines, and fleets available. The system proposes repair windows, books technicians, and triggers parts procurement across planning and shop-floor systems. The result is less unplanned downtime, lower maintenance cost, and higher throughput.

What - Process overview



Current challenges

- Condition data is fragmented across historians, controllers, and logs; issues are found late during manual rounds
- Maintenance is calendar-based or reactive; schedules fight with production plans, causing missed windows and expediting
- Root-cause analysis and recommended actions vary by technician; knowledge is trapped in notes, not reused
- Parts availability is uncertain; incorrect substitutions or late deliveries extend outages
- Handoffs between maintenance, production, and procurement require emails and spreadsheets, creating delays and errors



Frontier solution

Process description

- Continuously ingest sensor streams and events per asset; cleanse and align with maintenance history and operating context.
- Detect anomalies and predict failure risks with confidence and remaining-useful-life estimates; rank by impact on throughput and safety.
- Propose work orders with recommended actions, required parts, skills, and estimated durations; simulate candidate time windows against the production plan.
- Reserve technicians and tools, check parts stock and lead times, and place replenishment or expedited orders when needed.
- Orchestrate execution, confirm completion with digital checklists, and learn from outcomes to improve models and playbooks.

Solution description

Agents collaborate via events, share context, and hand off to humans for final sign-off:

- **Telemetry & Context Agent** aggregates IoT signals, events, and utilization; enriches with asset master data and maintenance history.
- **Anomaly & Failure Prediction Agent** detects deviations, estimates remaining useful life, and prioritizes risks by cost of failure.
- **Planner & Scheduling Agent** proposes work orders, aligns windows with production or route plans, and books technicians and tools.
- **Parts & Procurement Agent** verifies spares, selects approved substitutes, places purchase orders, and tracks supplier confirmations.
- **Execution & Learning Agent** guides the job with checklists, captures findings, and updates models, playbooks

Why – Impact¹



Business Outcomes²

Cost savings:

- 15–25% reduction in unplanned downtime from earlier detection and coordinated scheduling
- 10–20% lower maintenance cost through condition-based work and fewer emergency call-outs

Growth

- 2–5 percentage-point increase in asset availability and line throughput
- Higher on-time delivery and service reliability for fleets and plants



Roles addressed

- Reliability and maintenance leadership.
- Plant operations or fleet operations management.

How – Underlying foundation¹

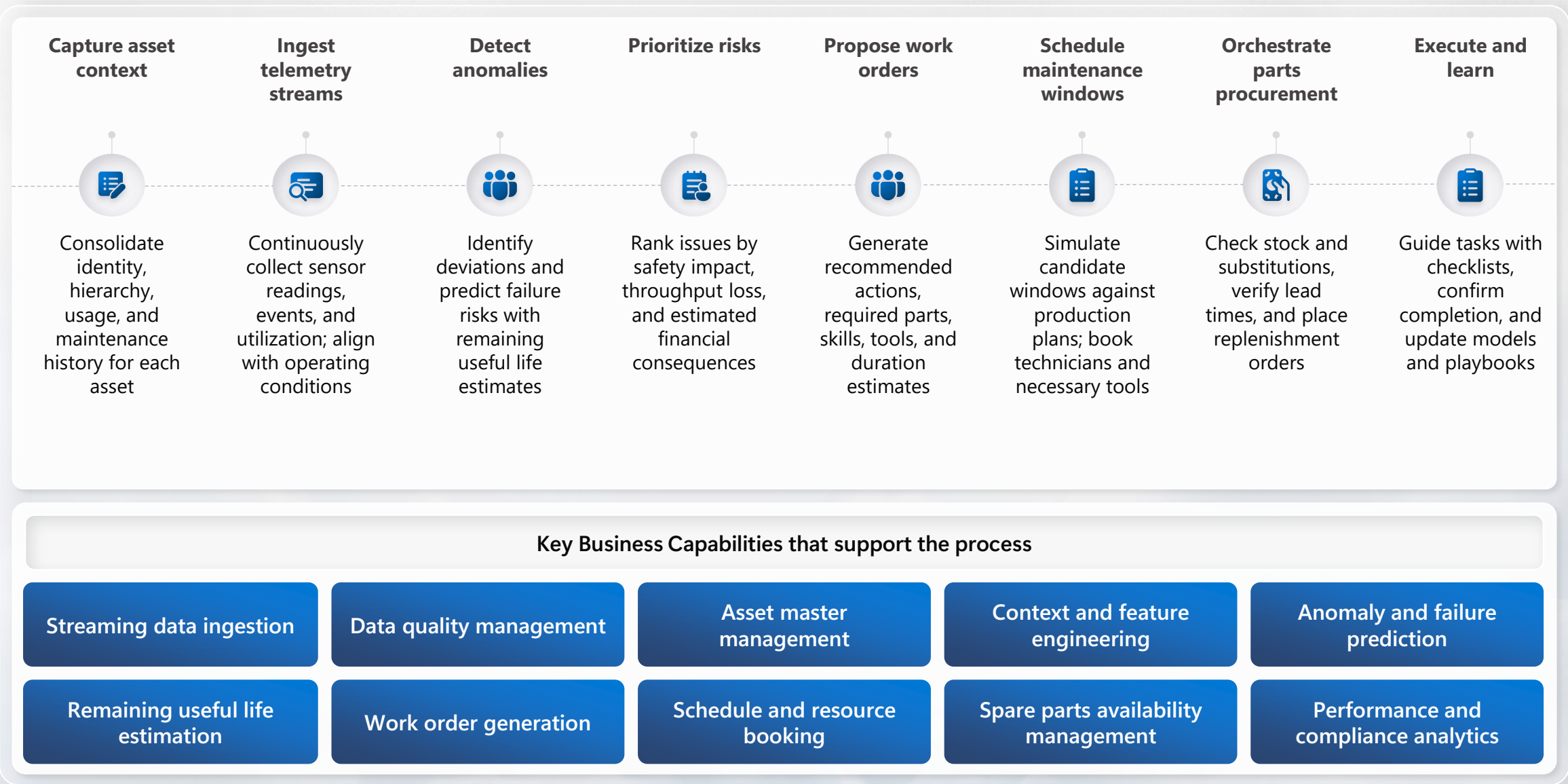
Data to integrate

- Sensor telemetry, control system events, and utilization by asset and line
- Asset master data, maintenance history, bills of materials, standard job plans, and safety procedures.
- Parts inventory, supplier lead times, purchase orders, production schedules.

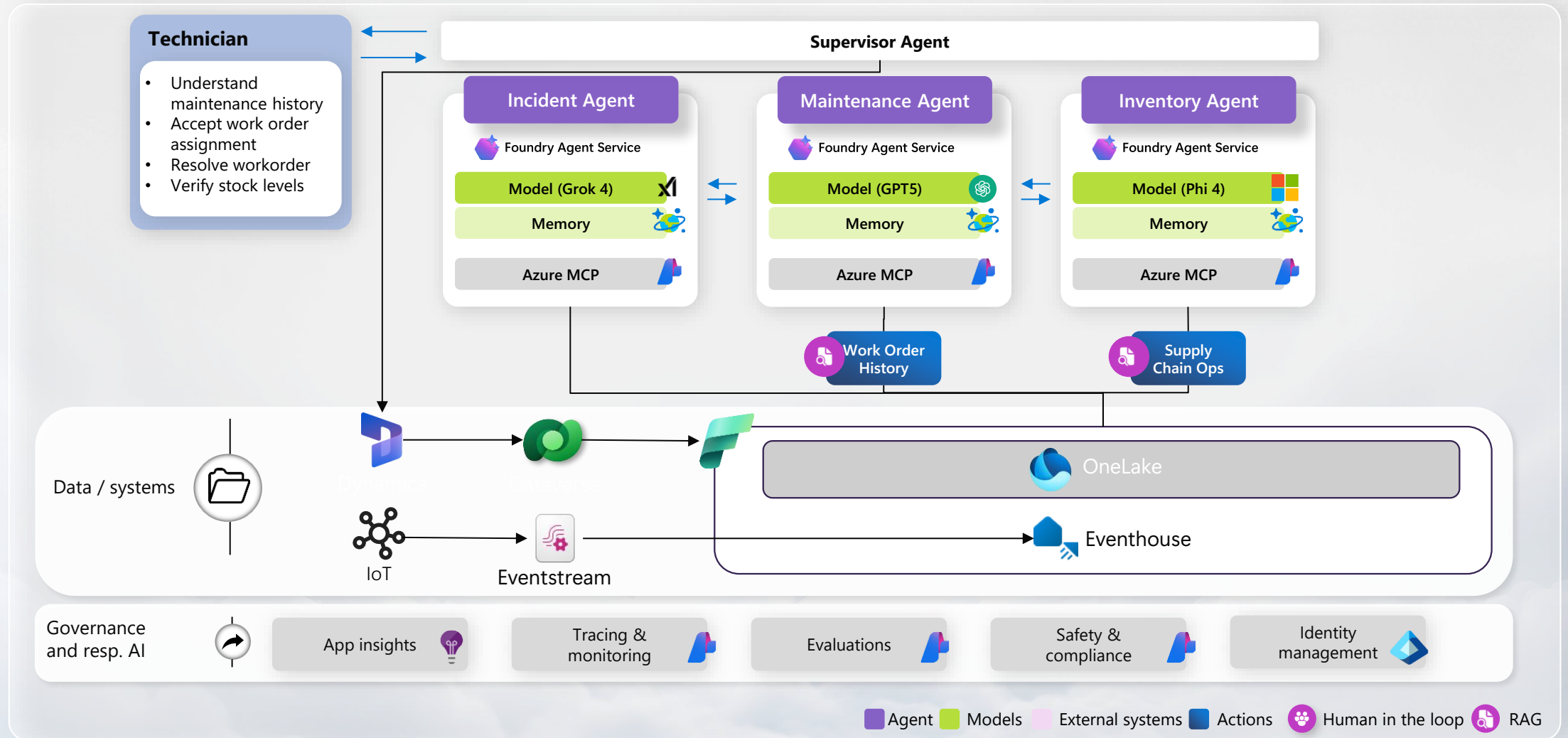
Technical patterns

- Streaming ingestion with feature store and probabilistic anomaly and failure prediction, including remaining-useful-life estimates.
- Event-driven orchestration that connects maintenance management, production planning, and procurement systems with human-in-the-loop approvals

2. What does the maintenance process look like?



Agentic Predictive maintenance, asset orchestration, plant operations





Knowledge mgmt. and
preservation

3. Knowledge mgmt. and preservation

Unify scattered documents, source code, Computer-Aided Design models, Product Lifecycle Management records, and service manuals into a governed, searchable hub. Chat over the full corpus—including legacy systems and file shares—so teams can find answers, reuse validated solutions, and preserve institutional know-how beyond retiring experts. The outcome is faster engineering work, fewer defects, and consistent service quality.

What - Process overview



Current challenges

- Knowledge is fragmented across network drives, email, issue trackers, Computer-Aided Design vaults, and Product Lifecycle Management; search is slow and incomplete
- Critical expertise lives with a few veterans; departures or supplier changes cause “black holes” in design rationale and service procedures
- Versions, approvals, and entitlements are unclear; teams risk using outdated drawings, code, or manuals
- Legacy systems are hard to access; scanned documents and mixed formats block reliable retrieval
- Rework and duplicate effort proliferate; field fixes and lessons learned do not flow back into standards



Frontier solution

Process description

- Ingest and continuously synchronize documents, drawings, bills of materials, change orders, source code, service manuals, test reports, and field notes from modern and legacy systems
- Normalize and enrich with optical character recognition, metadata, part numbers, variants, and lineage to specific models and components.
- Govern access with roles and need-to-know policies; track versions, approvals, and supersessions
- Enable chat over the corpus to answer “how/why/where” questions with citations, pointing to the authoritative file and version
- Capture new insights (for example, tribal knowledge, field fixes) into structured notes and link them to affected parts and procedures

Solution description

Agents collaborate via events, share context, and hand off to humans for final sign-off:

- **Ingestion & Enrichment Agent** connects to repos/file shares/CAD-PLM/code; runs OCR, extracts metadata, de-dups.
- **Entitlement & Governance Agent** enforces access, export controls, project limits; tracks versions and audits.
- **Search & Chat Orchestrator** retrieves across formats, cites sources, routes to the right version and related assets.
- **Lessons-Learned Curator** converts notes/emails/tickets into reusable knowledge linked to parts, procedures, failures.
- **Quality & Compliance Agent** flags outdated/conflicting content, missing approvals, and prompts standard updates.

Why – Impact¹



Business Outcomes²

Cost savings:

- 15–25% reduction in engineering search and rework time by finding the right file and rationale quickly
- 10–20% fewer defects and late changes through reuse of validated designs and procedures

Growth

- Shorter time-to-quote and time-to-design for variants, enabling faster responses to customer requests
- Higher customer satisfaction from consistent, accurate service guidance



Roles addressed

- Head of engineering or product
- After-sales and service engineering leadership

How – Underlying foundation¹

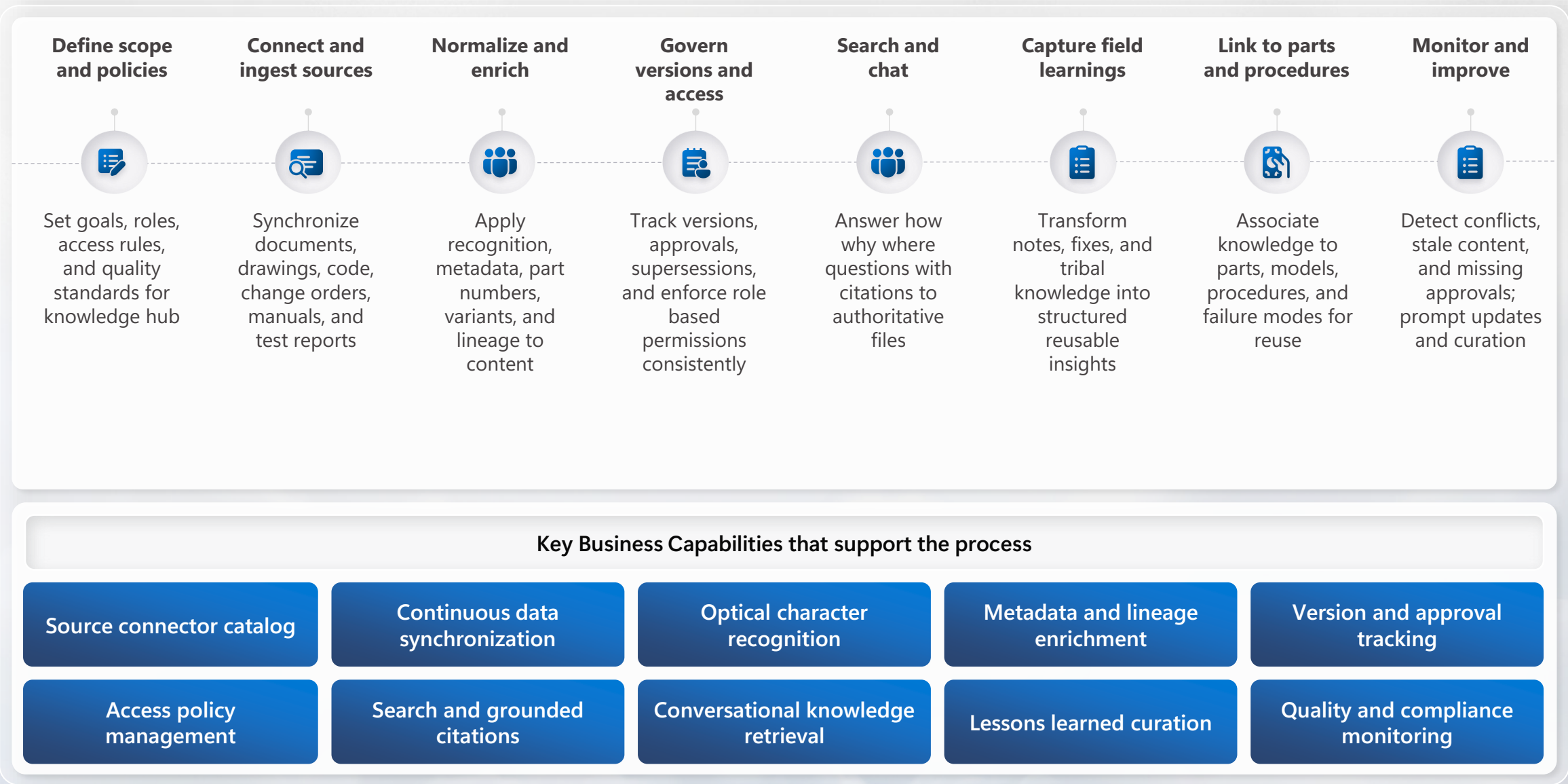
Data to integrate

- Engineering artifacts: Computer-Aided Design models and drawings, Product Lifecycle Management records, bills of materials, change orders, and test reports.
- Source code, build and release notes, issue and ticket systems for tools, diagnostics, and embedded software.

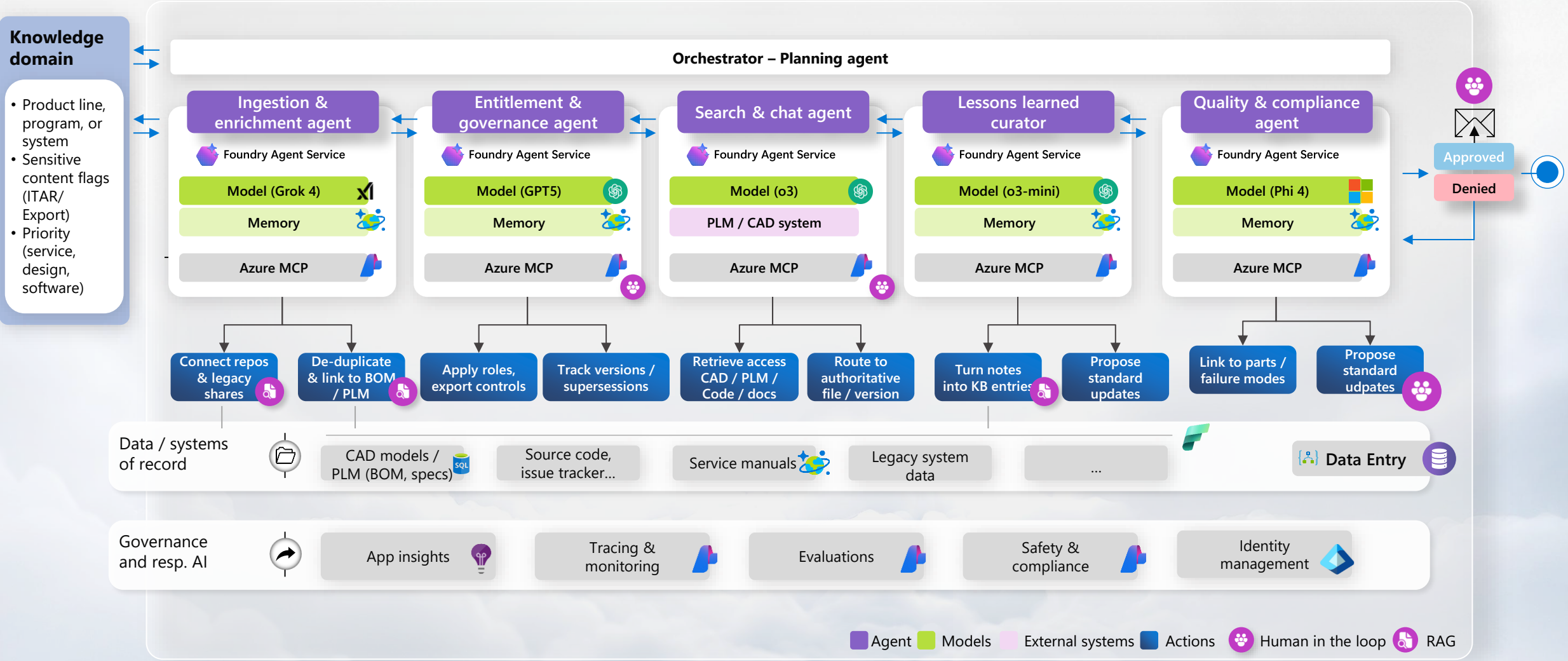
Technical patterns

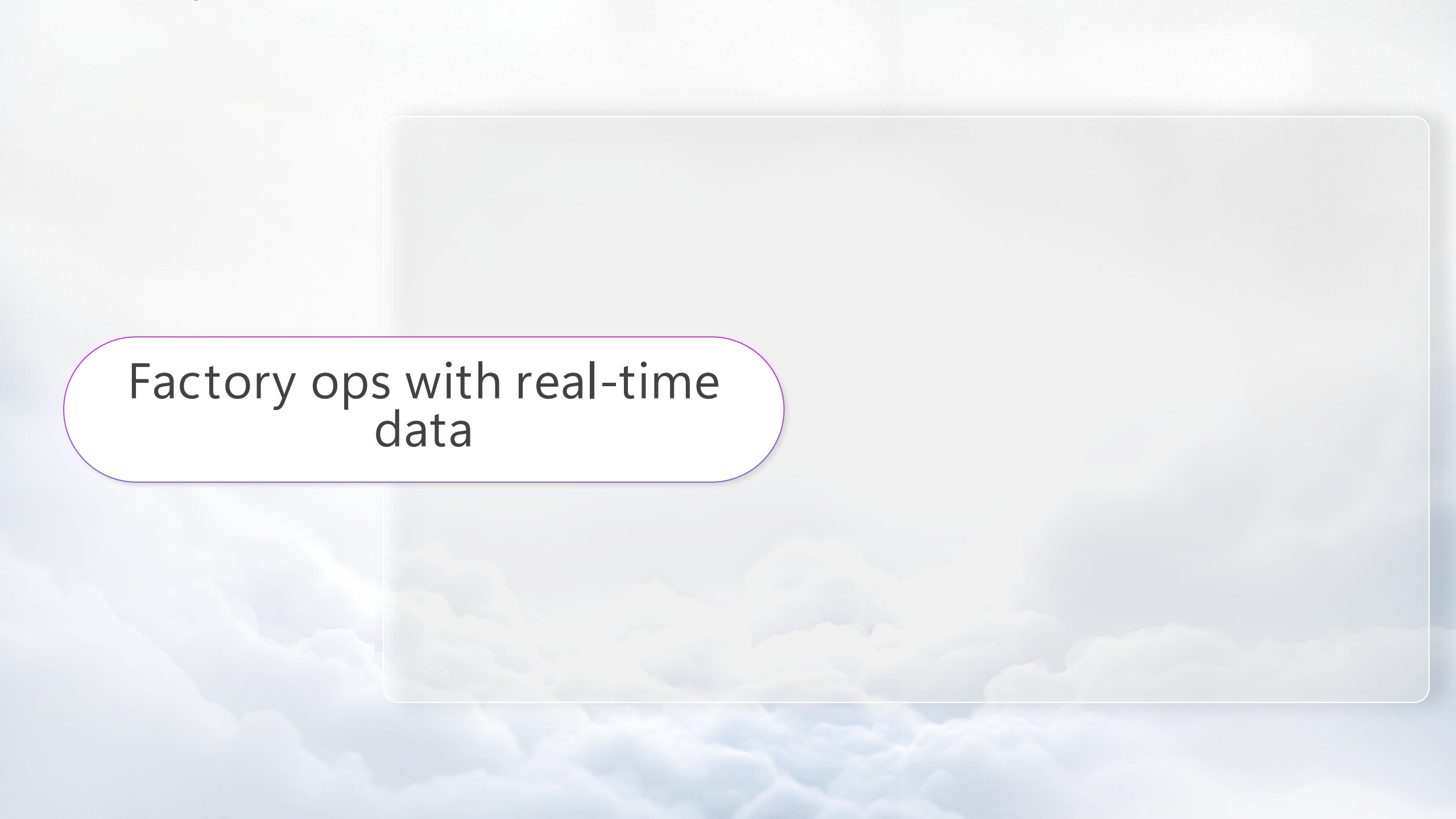
- Retrieval and grounded chat over repositories with policy-aware access, version control linkage, and citations
- Document and code ingestion pipeline with metadata/part-number enrichment, de-duplication, lineage tracking, and human-in-the-loop curation

3. What does the knowledge mgmt. and preservation process look like?



3. Example of agentic knowledge mgmt. and preservation





Factory ops with real-time data

4. Factory ops with real-time data

Agents coordinate Manufacturing Execution Systems, Advanced Planning and Scheduling, and Warehouse Management Systems to keep line cadence and flow—balancing work across stations, planning changeovers, kitting and delivering materials just in time, and pacing energy use. The result is fewer buffers and idle minutes, higher Overall Equipment Effectiveness, and greater throughput without sacrificing quality or safety.

What - Process overview



Current challenges

- Plans drift when machines, labor, or materials change; rescheduling takes hours and relies on manual spreadsheets.
- Changeovers and setups are booked on fixed calendars; actual demand, tooling, and upstream readiness are ignored.
- Materials arrive early or late; kitting errors or missing substitutes cause starved stations and rework.
- Energy-intensive steps peak at the wrong times; utilities costs rise and thermal constraints are breached.
- Exceptions (scrap spikes, late inbound, operator absence) lead to firefighting and uncoordinated workarounds.



Frontier solution

Process description

- Continuously ingest orders, routings, work-in-process, machine states, quality yields, labor availability, and energy/utility status.
- Build a live plan for lines and cells; re-balance work when constraints shift to protect line cadence and service targets.
- Schedule changeovers based on demand, setup matrices, tooling, and upstream readiness; pre-stage kits and verify substitutes.
- Orchestrate material calls, Automated Guided Vehicle missions, and dock appointments to arrive just in time.
- Pace energy-intensive steps to off-peak windows where feasible; monitor limits and adjust sequences safely.
- Surface proposed changes with impact on throughput, scrap, and cost; execute approved updates and learn from outcomes.

Solution description

Agents collaborate via events, share context, and hand off to humans for final sign-off:

- Line Flow Orchestrator** balances stations/cells, re-sequences jobs, and protects cadence
- Changeover & Setup Planner** schedules setups from demand, matrices, tooling readiness, and skills; minimizes lost minutes
- Materials Kitting & Delivery Agent** builds kits, verifies substitutes, and triggers JIT delivery/AGV missions to point of use
- Energy Pacing & Utilities Agent** staggers energy-intensive ops, enforces safety limits, proposes off-peak schedules
- Exception & Recovery Agent** detects disruptions (scrap bursts, late inbound, absenteeism) and proposes recovery playbooks across production/warehouse

Why – Impact¹



Business Outcomes²

Cost savings:

- 10–20% reduction in work-in-process and buffers through tighter flow and just-in-time material delivery
- 12–25% fewer changeover and setup losses via demand-aware sequencing and pre-staging

Growth

- +3–7 percentage-points in Overall Equipment Effectiveness from higher availability and performance
- 5–10% throughput uplift at similar staffing and footprint through dynamic balancing and faster recovery



Roles addressed

- Plant operations or production control leadership
- Warehouse and intralogistics management

How – Underlying foundation¹

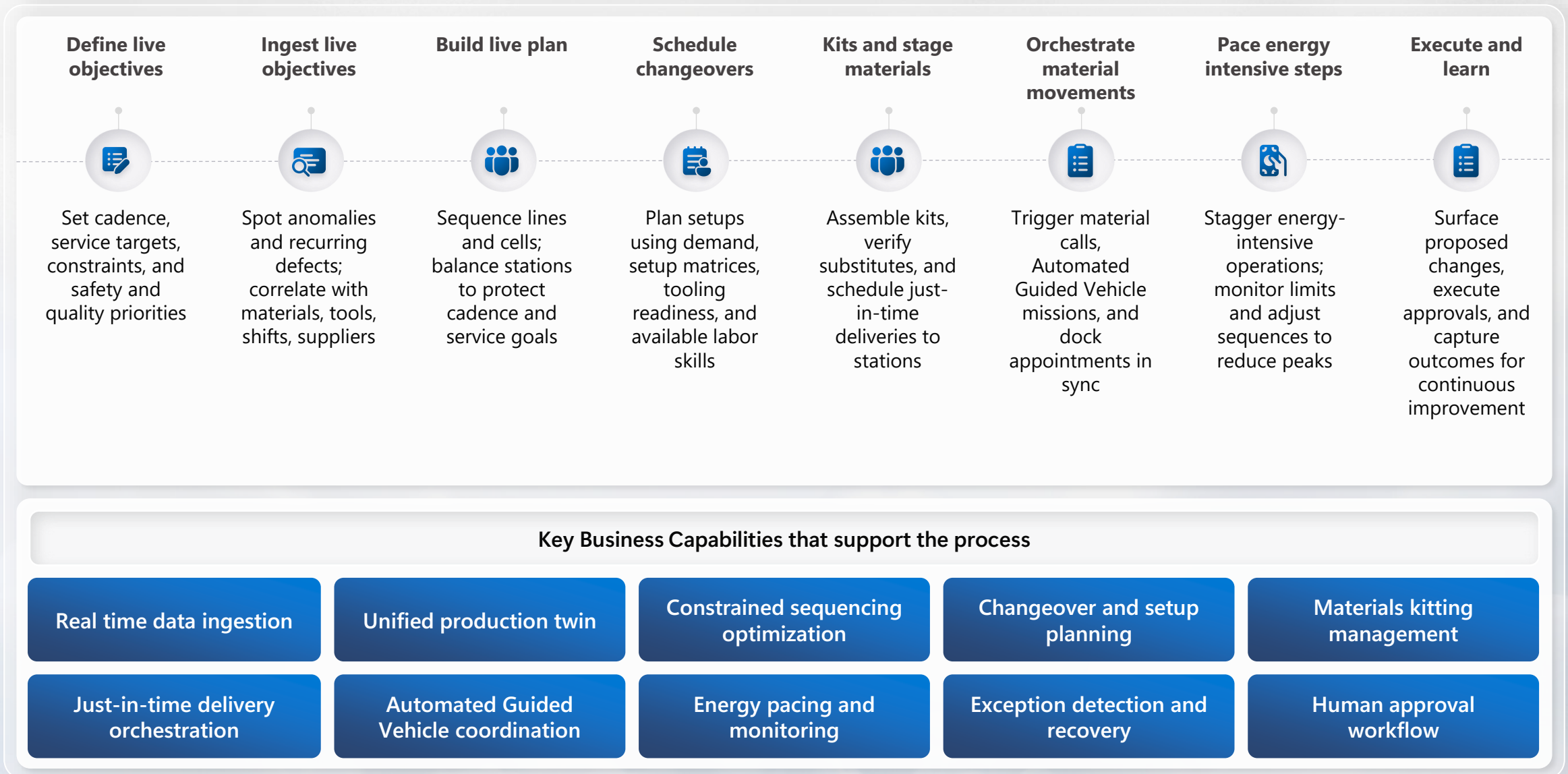
Data to integrate

- Orders, routings, bills of materials, work-in-process, and quality yields by station and line
- Machine states, alarms, and maintenance windows with labor rosters and skills
- Inventory positions, kitting status, locations and missions

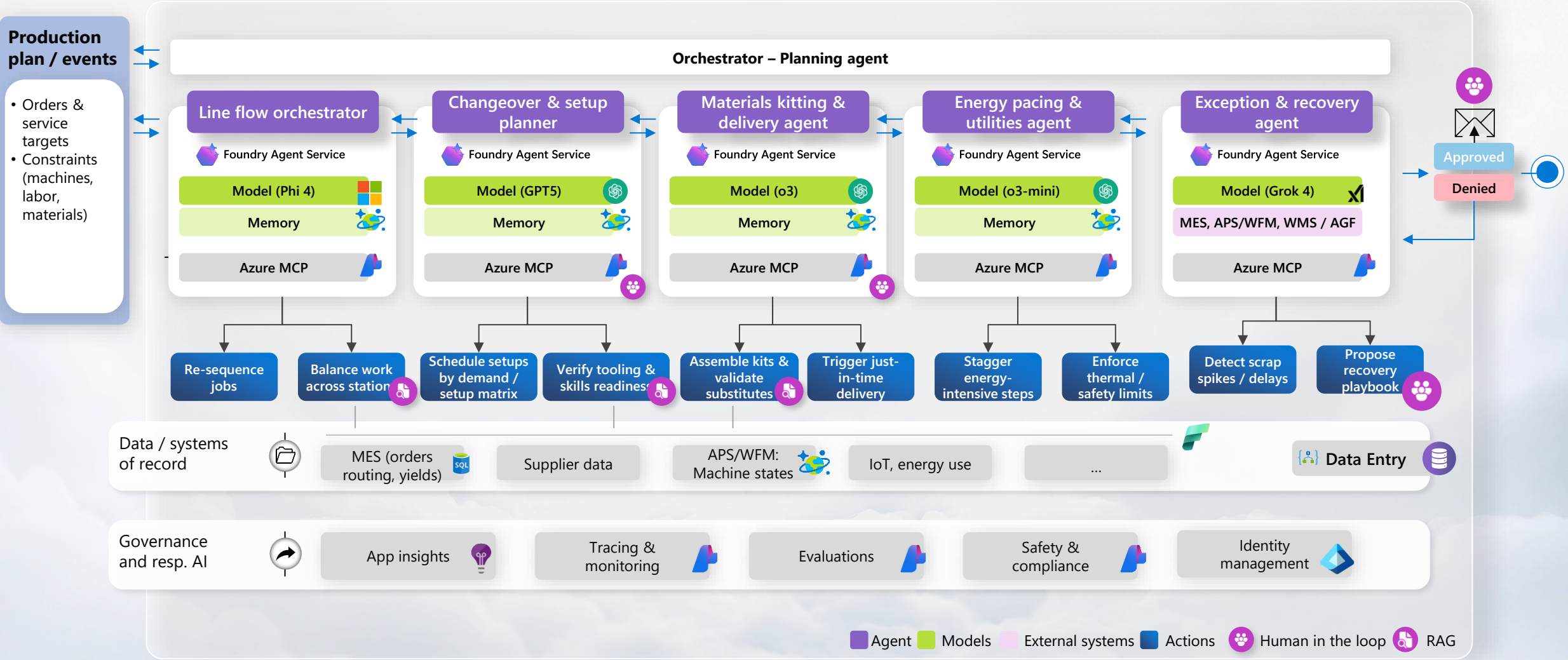
Technical patterns

- Event-driven digital-twin state with constrained optimization for sequencing, changeovers, and material flow
- Multi-agent orchestration across MES, Advanced Planning and Scheduling, and Warehouse Management Systems with human-in-the-loop approvals

4. What does the factory ops with real-time data process look like?



4. Example factory ops with real-time data management



Requirements engineering

5. Engineering-to-order

Requirements-driven engineering uses agents to mine customer, regulatory, and field signals to draft and trace requirements, generate specifications and tests, run “design-for-X” checks (for example, design for manufacturing, assembly, service, safety), and flag change impacts. Teams shorten engineering cycles, reduce rework, and improve launch quality with clear lineage from need to design, verification, and release.

What - Process overview



Current challenges

Inputs are scattered across customer feedback, service tickets, standards, and past projects; requirements are incomplete or inconsistent.

Traceability from requirements to design, test cases, and approvals is manual; changes ripple unpredictably and are discovered late.

Verification plans lag behind design; tests miss real-world failure modes from service and warranty data.

“Design-for-X” reviews are ad-hoc; manufacturability, assembly, serviceability, cost, and safety issues surface during pilot builds.

Compliance evidence is hard to assemble for stage-gates and audits.



Frontier solution

Process description

- Continuously ingest signals from customers, field service, warranty, regulatory updates, and competitor benchmarks.
- Draft and de-duplicate requirements with clear acceptance criteria, priority, and measurable targets; map to affected systems and components.
- Auto-generate specifications and linked test cases; align verification plans with risks and standards.
- Run design-for-X checks (manufacturing, assembly, service, cost, safety) against current models and bills of materials; propose fixes.
- When a change occurs, analyze upstream and downstream impacts; update requirements, tests, and evidence packs for stage-gates.

Solution description

Agents collaborate via orchestrator agent and humans in the loop:

- **Signal & Voice-of-Customer Agent** mines reviews, service logs, warranty claims, bulletins, competitor data
- **Requirements & Traceability Agent** drafts requirements w/ acceptance criteria, links to systems/components/owners
- **Specification & Test Generator** creates specs and verification plans; synthesizes test cases/datasets aligned to standards/risks
- **Design-for-X Checker** assesses manufacturability, assembly, serviceability, cost, safety against rules/history
- **Governance Orchestrator** simulates change ripple effects, updates artifacts, gathers approvals, compiles audit-ready evidence

Why – Impact



Business Outcomes

Efficiency

- 20–30% faster requirement-to-design cycle through automated drafting, linking, and verification planning
- 15–25% reduction in engineering rework by catching manufacturability and compliance issues earlier

Growth

- Higher launch reliability and customer satisfaction from designs aligned to real-world use and service data
- Shorter time-to-market for variants through reusable requirements, tests, and evidence packs



Roles addressed

- Systems engineering and product management leaders
- Component design leads and verification and validation managers.

How – Underlying foundation

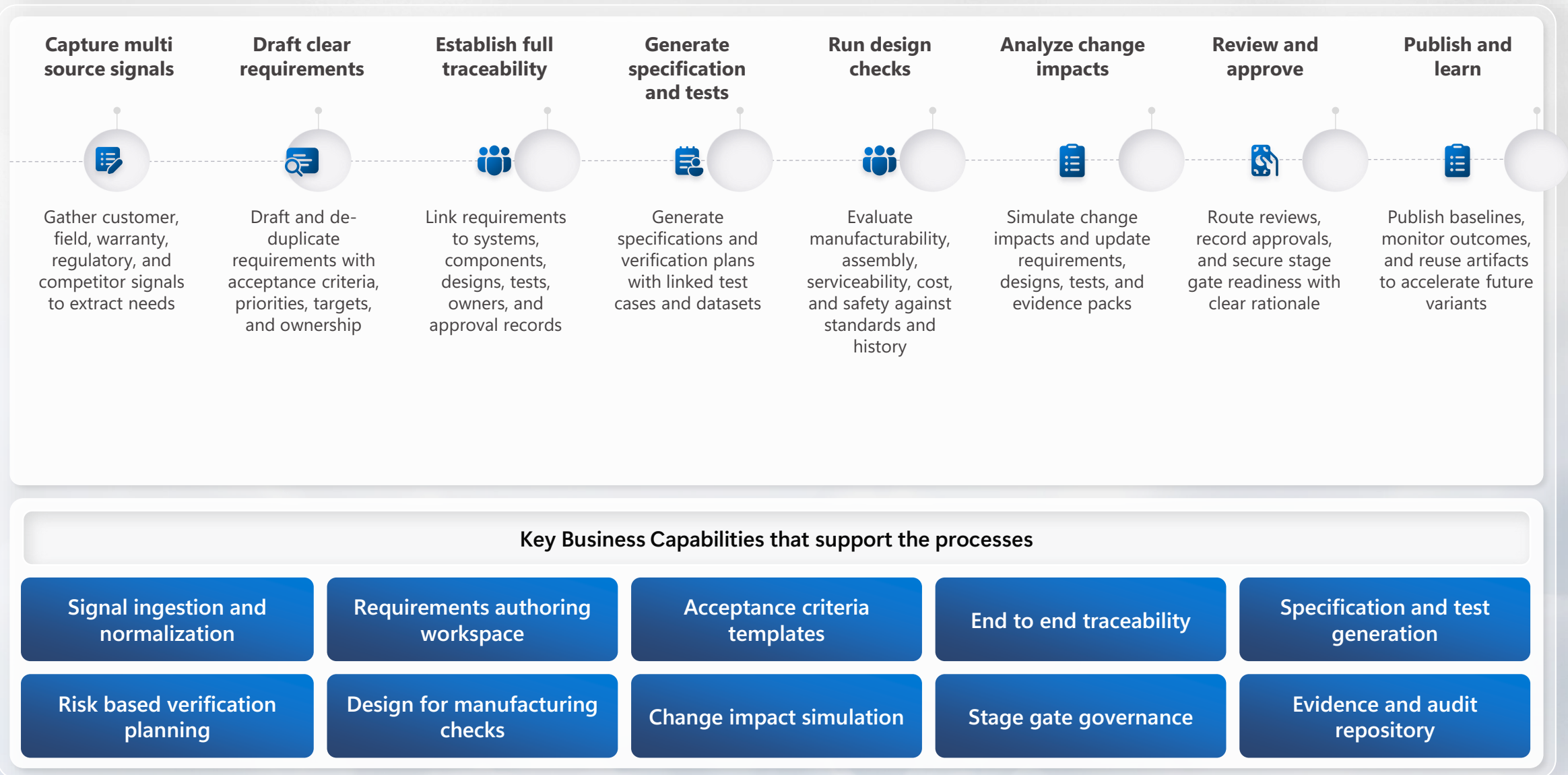
Data integration

- Customer and field signals: service tickets, warranty claims, returns, usage modes
- Standards, regulations, safety bulletins, and competitor or benchmark specs
- Engineering artifacts: requirements and change logs, models and drawings, bills of materials, and historical test results

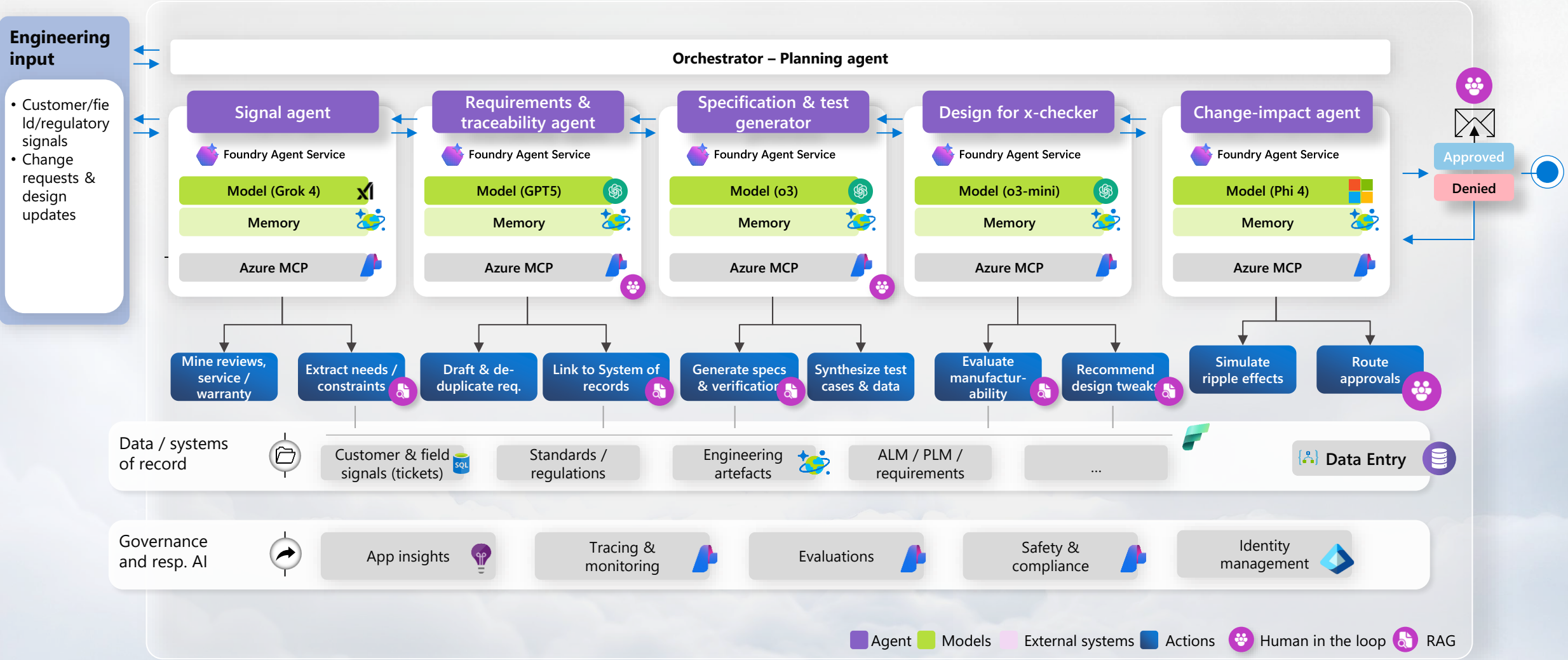
Technical patterns

- Retrieval and grounded reasoning over multi-source engineering and policy data
- Tool-using agent orchestration for rule-based design-for-X checks and change-impact simulation with human-in-the-loop approvals and auditable stage-gate evidence

5. What does the engineering-to-order process look like?



5. Example architecture for agentic engineering-to-order process





Thank you